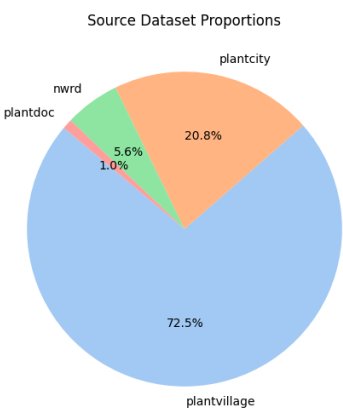
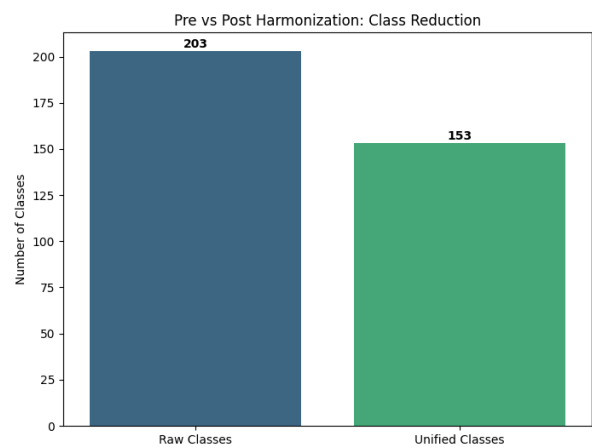


ZARI.ai Dataset Harmonization & Audit Report

This report summarizes the data engineering phase of the ZARI.ai project. The objective was to harmonize raw agricultural datasets, enforce strict crop-disease boundaries, and prepare stratified splits for model training.

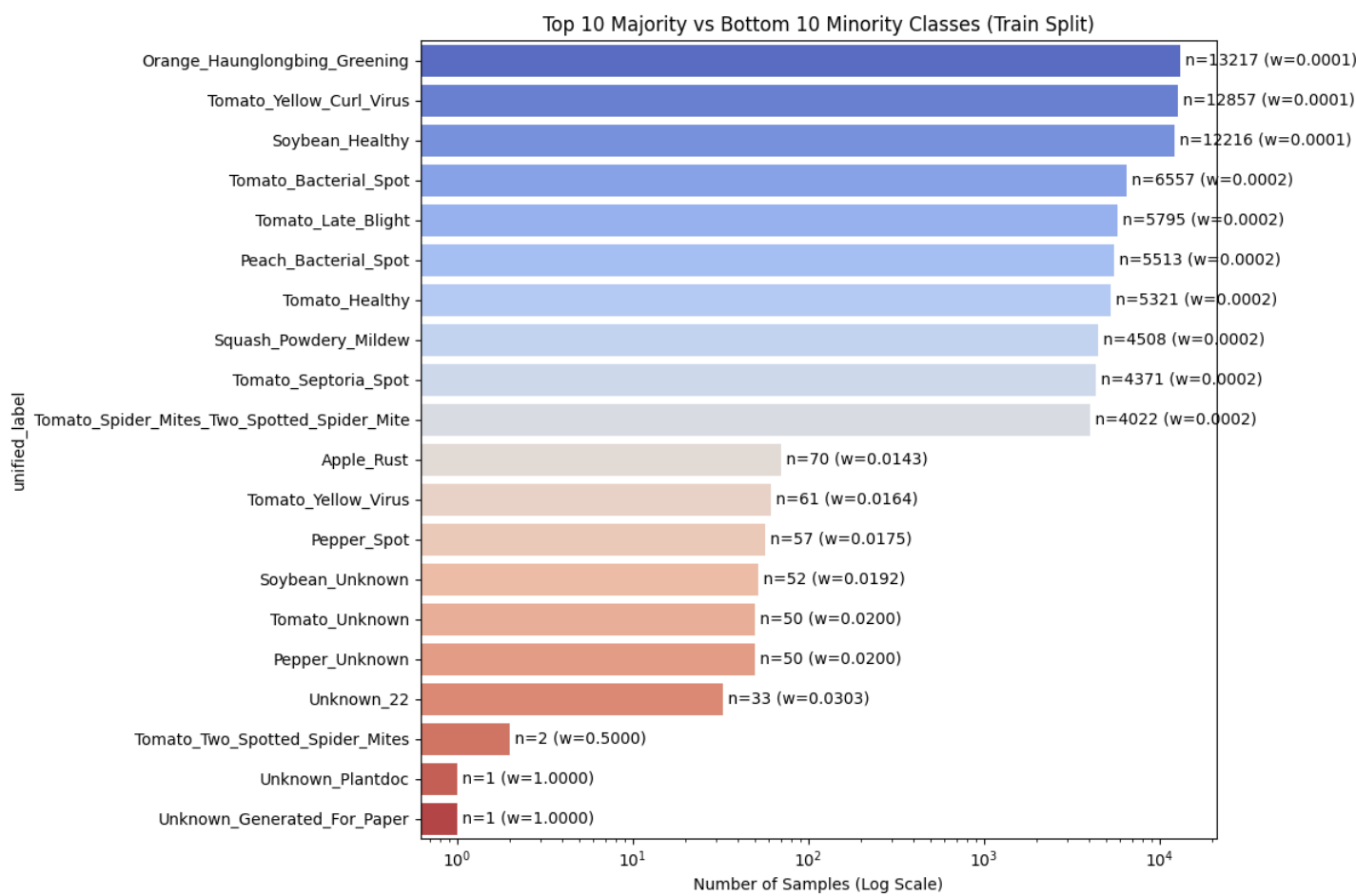
Summary Statistics

Total Indexed Samples:	251,215
Classes (Raw -> Unified):	203 -> 153
Train / Val / Test Splits:	202,753 / 24,006 / 24,456
Extreme Imbalance Ratio:	13,217 : 1 (Orange_Haunglongbing_Greening vs Unknown_Generated_For_Paper)



Class Imbalance & Stratification Audit

The agricultural datasets exhibit extreme real-world class imbalance, reaching up to 13k:1 ratios. To mitigate this, `ml_pipeline/03_dataset.py` implements a `WeightedRandomSampler`. The stratification algorithm also successfully maintained the 80/10/10 split across all 138 classes, ensuring no data leakage.



Sign-Off & Next Steps

Data integrity verified. The dataset is fully harmonized, stratified, and balanced via sampling weights. The pipeline is cleared to proceed to Model Architecture (`ml_pipeline/04_model.py`).